Institute of Technology of Cambodia
Department of Applied Mathematics and Statistics

Present by Group 5

MEDICAL APPOINTMENT NO-SHOWS

Lecturer: Dr. HAS Sothea(Course)
Ms. UANN Sreyvi(TP)

GROUP MEMBERS

SORN Sreynatt
e20210292

ROEUN Sovandeth
e20211022

SAN Kimheang
e20210323

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1. INTRODUCTION

WHAT IS A MEDICAL APPOINTMENT NO-SHOW?

A no-show occurs when a patient does not attend a scheduled medical appointment and provides no prior notice or cancellation.
Key Characteristics:

- Patient is absent at the scheduled time
- No advance communication to the healthcare facility
- Appointment slot goes unused

01

PROBLEM

- Missed appointments in public healthcare waste limited resources and delay care for millions.

Impact:

- Financial losses
- Longer wait times
- Worse health outcomes
- Operational inefficiency

02

OUR APPROACH

We use advanced analytics—PCA, Clustering, GMM, and Correspondence Analysis—to uncover hidden patterns and identify high-risk patients.

03

GOAL

To enable data-driven strategies that reduce no-shows and improve healthcare access in Brazil's SUS.

2. DATASET OVERVIEW

FIRST 5 ROWS OF MEDICAL APPOINTMENT DATASET

	PatientId	AppointmentID	Gender	ScheduledDay	AppointmentDay	Age	Neighbourhood	Scholarship	Hipertension	Diabetes	Alcoholism	Handcap	SMS_received	No-show
0	2.987250e+13	5642903	F	2016-04-29T18:38:08Z	2016-04-29T00:00:00Z	62	JARDIM DA PENHA	0	1	0	0	0	0	No
1	5.589978e+14	5642503	M	2016-04-29T16:08:27Z	2016-04-29T00:00:00Z	56	JARDIM DA PENHA	0	0	0	0	0	0	No
2	4.262962e+12	5642549	F	2016-04-29T16:19:04Z	2016-04-29T00:00:00Z	62	MATA DA PRAIA	0	0	0	0	0	0	No
3	8.679512e+11	5642828	F	2016-04-29T17:29:31Z	2016-04-29T00:00:00Z	8	PONTAL DE CAMBURI	0	0	0	0	0	0	No
4	8.841186e+12	5642494	F	2016-04-29T16:07:23Z	2016-04-29T00:00:00Z	56	JARDIM DA PENHA	0	1	1	0	0	0	No

DATA SOURCE & SCOPE

- 10,527 medical appointments from Brazilian public healthcare (SUS) in May 2016
- 14 original variables covering demographics, appointment details, health conditions, and outcomes
- Geographic diversity: 81 different neighborhoods across Brazil

VARIABLES IN THE MEDICAL APPOINTMENT DATASET

1. PatientId – Unique patient ID
2. AppointmentID – Unique appointment ID
3. Gender – Patient gender (F/M)
4. ScheduledDay – Date/time appointment was scheduled
5. AppointmentDay – Appointment date
6. Age – Patient's age in years
7. Neighbourhood – Location of health facility
8. Scholarship – Enrolled in welfare program (0/1)
9. Hipertension – Hypertension diagnosis (0/1)
10. Diabetes – Diabetes diagnosis (0/1)
11. Alcoholism – Alcoholism diagnosis (0/1)
12. Handcap – Disability level (0-4)
13. SMS_received – Reminder SMS sent (0/1)
14. No-show – Missed appointment? (Yes/No)

DATA PREPROCESSING

DATA PREPROCESSING

- Missing Values Check – None found
- Duplicate Check – No duplicate appointments
- Invalid Age Removal – 1 record with age -1 removed
- → Final dataset: 110,526 records
- Date Format Conversion – ScheduledDay & AppointmentDay standardized to datetime

FEATURE ENGINEERING (NEW VARIABLES CREATED)

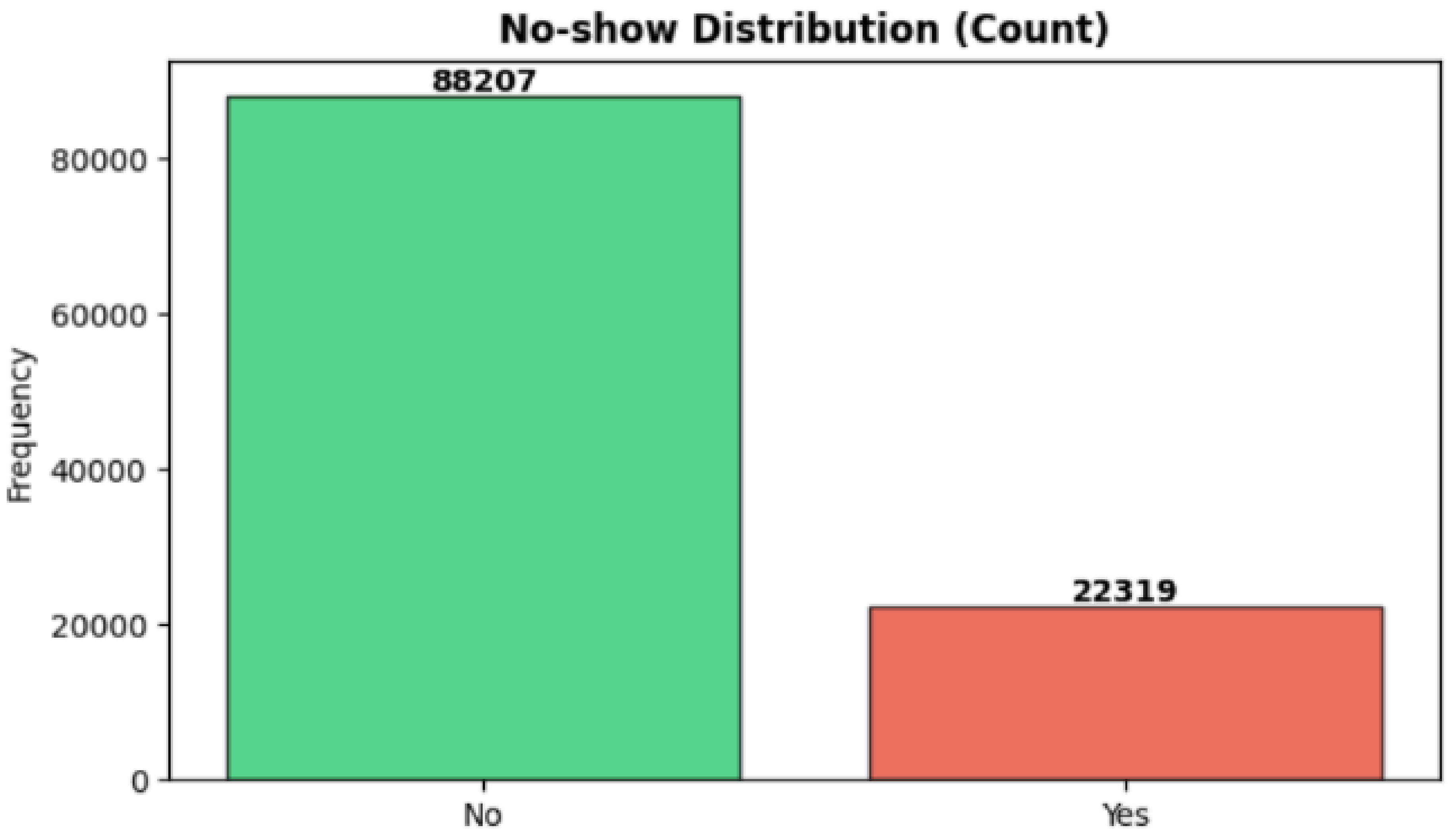
- AwaitingDays – Days between scheduling & appointment date
- AgeGroup – Child, Teen, Young Adult, Adult, Senior
- IsChild – 1 if age < 18, else 0
- IsSenior – 1 if age ≥ 60, else 0
- TotalConditions – Sum of health conditions (hypertension, diabetes, alcoholism, disability)
- HasChronicCondition – 1 if hypertension or diabetes, else 0
- HasMultipleConditions – 1 if TotalConditions ≥ 2, else 0
- NoShow_binary – 1 if no-show, 0 if attended
- WaitingCategory – Same-day, 1–7 days, 8–30 days, 31–90 days, 90+ days

3. EXPLORATORY ANALYSIS

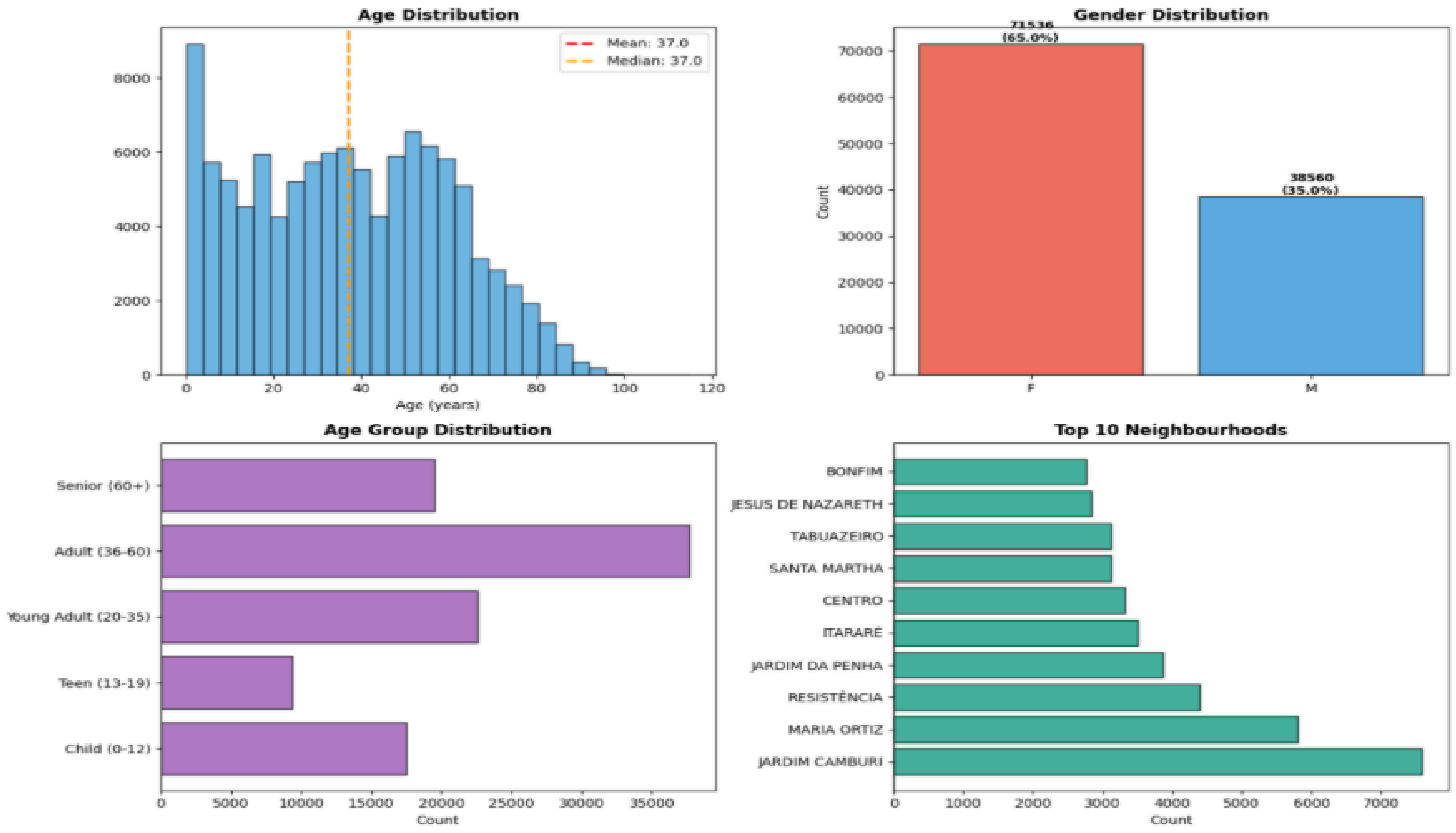
3.1. UNIVARIATE ANALYSIS

UNIVARIATE ANALYSIS

- Univariate analysis was used to examine the distribution of individual variables
- Datetime-based features were extracted to capture temporal patterns in patient attendance
- Appointment counts were analyzed across time periods to identify peak scheduling times
- These time-related features help explain and predict no-show behavior

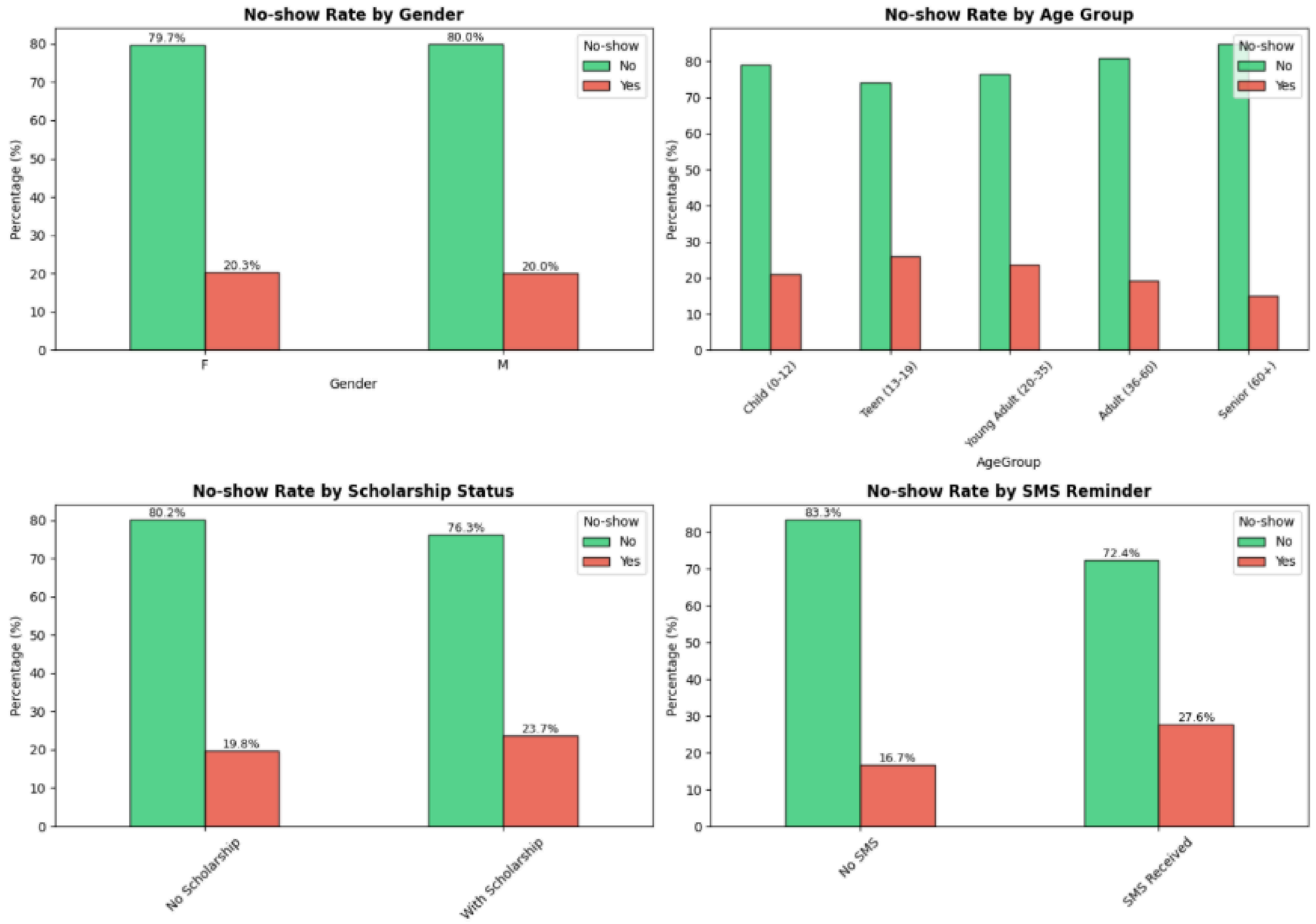


3.1. **UNIVARIATE ANALYSIS**



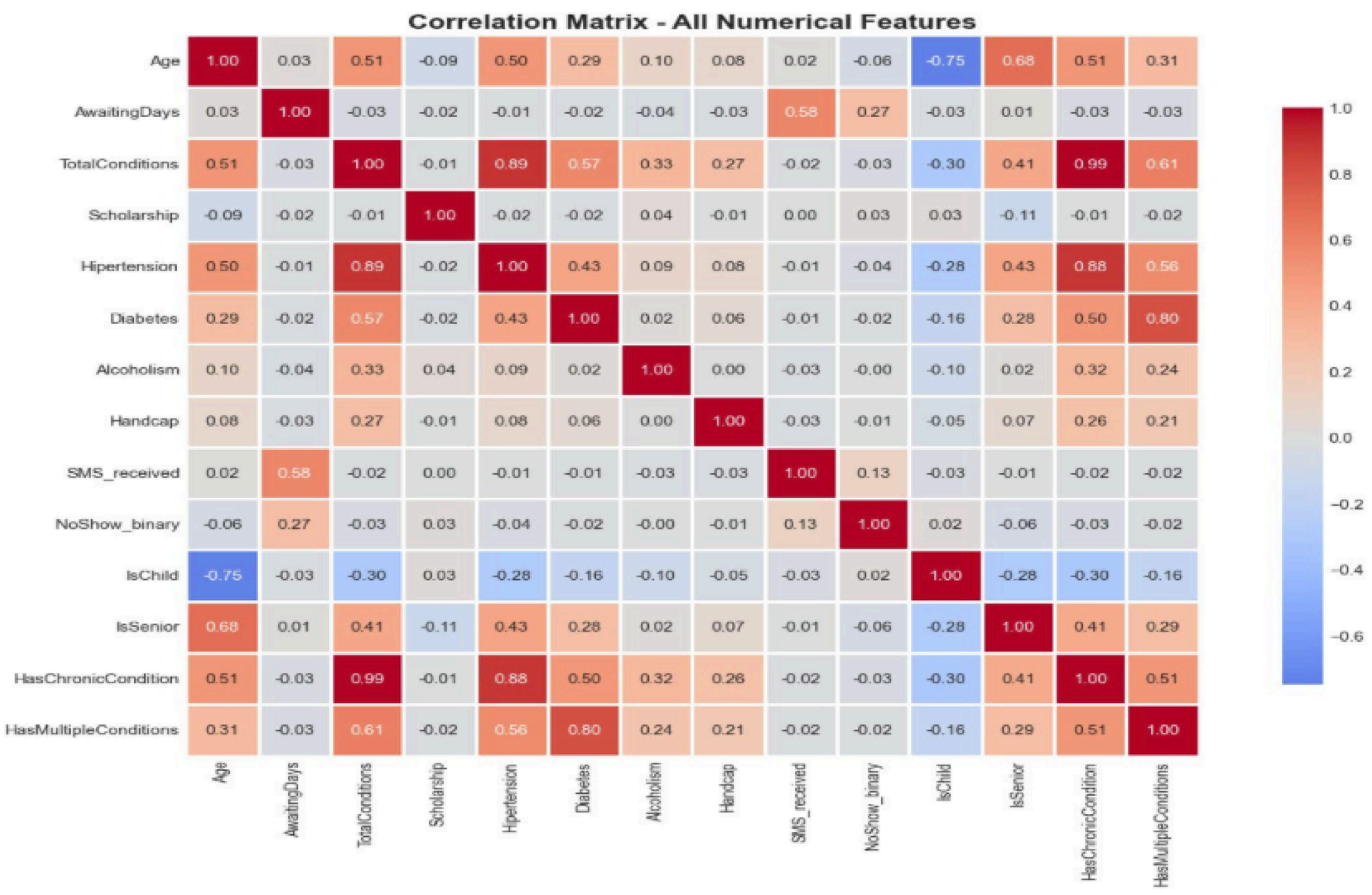
- Average patient age is 37 years (median = 37), indicating a balanced age distribution
- Age shows high variability (SD = 23), reflecting diverse age groups
- Females account for 65% of appointments, compared to 35% males
- Adults (36–60) form the largest age group, followed by young adults (20–35)
- Seniors and children are moderately represented, while teenagers are the smallest group

3.2. **BIVARIATE ANALYSIS**



- Gender
- Female no-show rate: 20.29%
- Male no-show rate: 19.95%
- Difference is minimal → gender has a weak effect
- Age Group
- Teenagers (13–19): 25.95% (highest)
- Young adults (20–35): 23.70%
- Children (0–12): 20.93%
- Adults (36–60): 19.08%
- Seniors (60+): 15.13% (lowest)
- Key Insight
- Age is a strong predictor of no-show behavior
- Younger patients are more likely to miss appointments
- Older patients show higher appointment reliability

3.2. **BIVARIATE ANALYSIS**

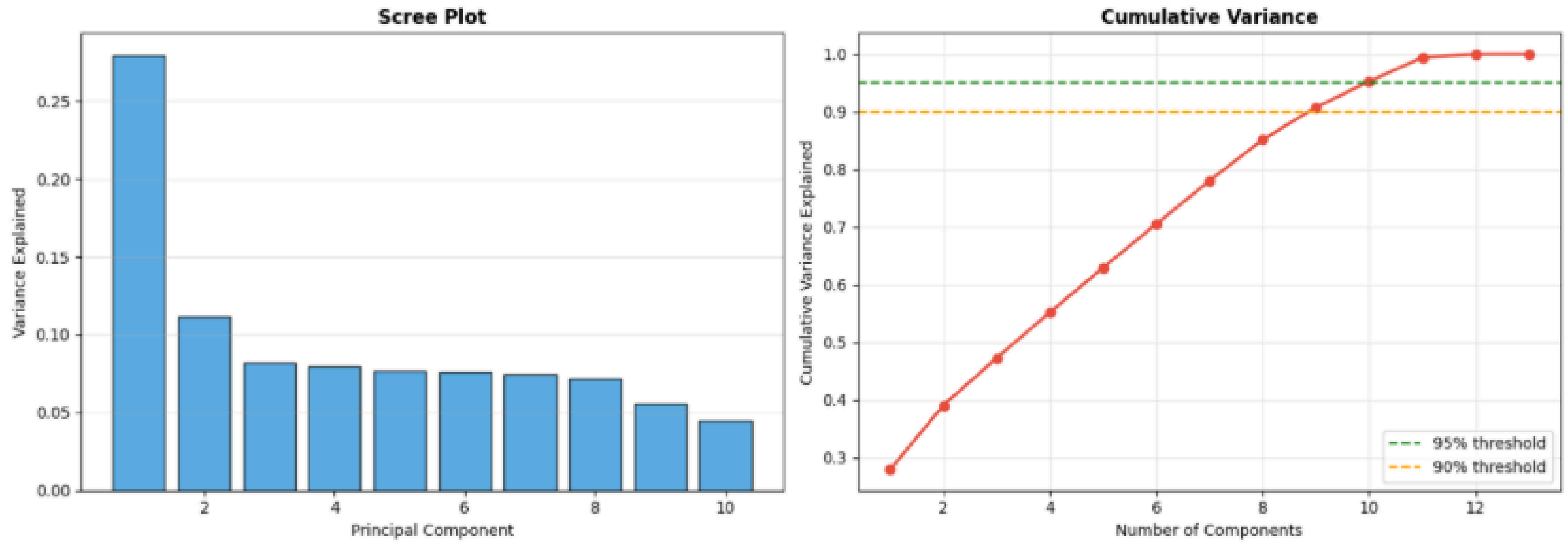


- AwaitingDays
- Small positive correlation with no-show (0.27)
- Longer waiting time slightly increases missed appointments
- Age & Health Factors
- Age, TotalConditions, HasChronicCondition, HasMultipleConditions → very weak correlation with no-show
- Health-related variables are highly correlated with each other, but not with no-show
- Age Flags
- IsChild and IsSenior strongly correlated with age
- Weak negative correlation with no-show → children and seniors are slightly more reliable
- SMS Reminders
- Small positive correlation (0.13)
- Likely reflects reminders being sent to high-risk patients, not causing no-shows
- Key Takeaway
- No single variable strongly predicts no-show behavior
- No-show risk is influenced by multiple small effects

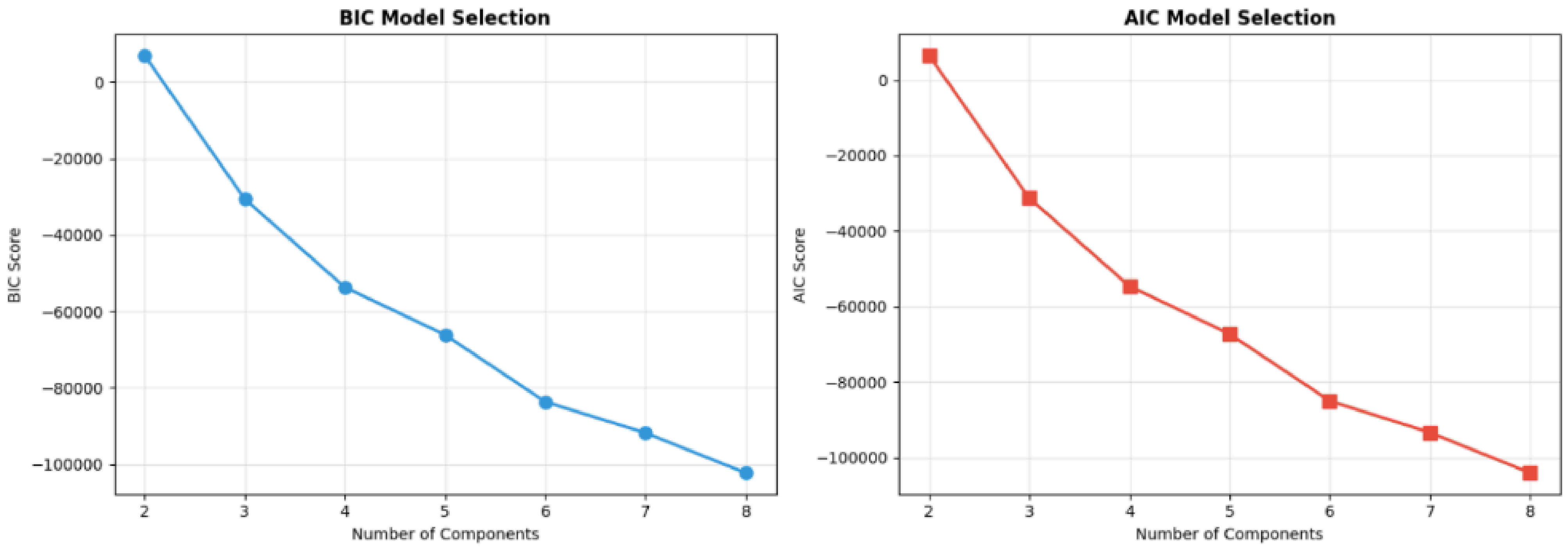
4. METHODOLOGY

4.1. PRINCIPAL COMPONENT ANALYSIS (PCA)

- PCA applied to standardized numeric features for dimensionality reduction
- Top 5 components explain ~63% of total variance
- PC1: 28%, PC2: 11%
- 10–11 components needed to capture 90% of variance
- Indicates moderate redundancy in the dataset
- PCA simplifies the data while retaining most information for modeling

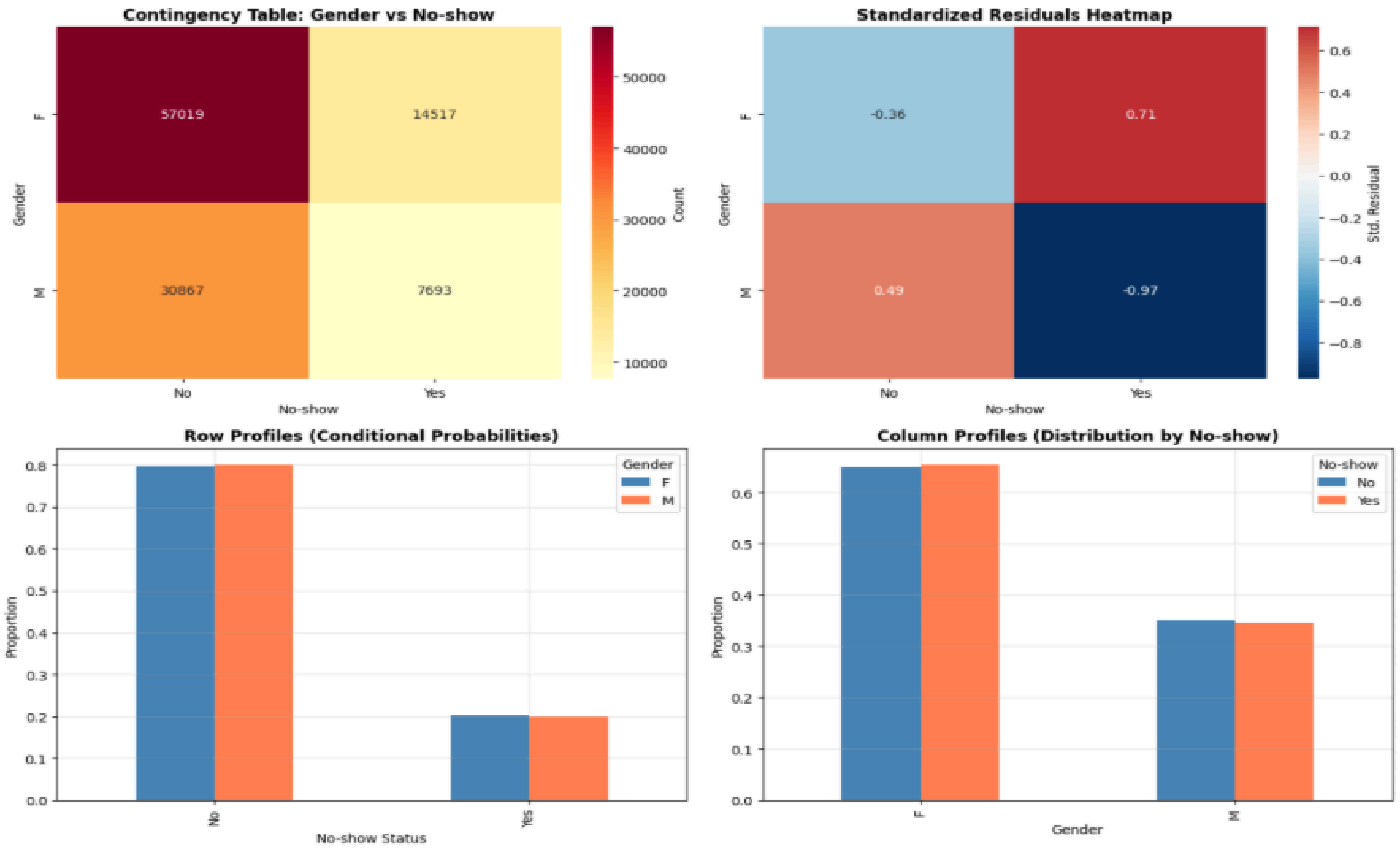


GAUSSIAN MIXTURE MODELS (GMM)



- GMM captures overlapping clusters, unlike K-means' hard assignments
- Identifies multiple patient profiles with shared characteristics
- Indicates no-show behavior is complex and not clearly separable
- Suggests no-shows are driven by subtle, combined patterns
- Provides a more flexible patient segmentation for analysis

CORRESPONDENCE ANALYSIS (CA)



- Very weak association between gender and no-show (χ^2 = 1.83, df = 1)
- Standardized residuals near zero → observed counts close to expected
- No-show rates almost identical: Females 20.3%, Males 19.9%
- Column profiles show no significant gender difference among no-shows
- Gender has minimal effect

MODEL SPECIFICATION

Based on data characteristics and analysis results, tree-based models such as XGBoost and Random Forest are recommended for no-show prediction. These models handle mixed features, non-linear relationships, and class imbalance effectively. Performance should prioritize recall and AUC-ROC to better identify no-show patients.

XGBoost Model

- The XGBoost model was trained on the prepared dataset with class balancing to address the no-show class imbalance.

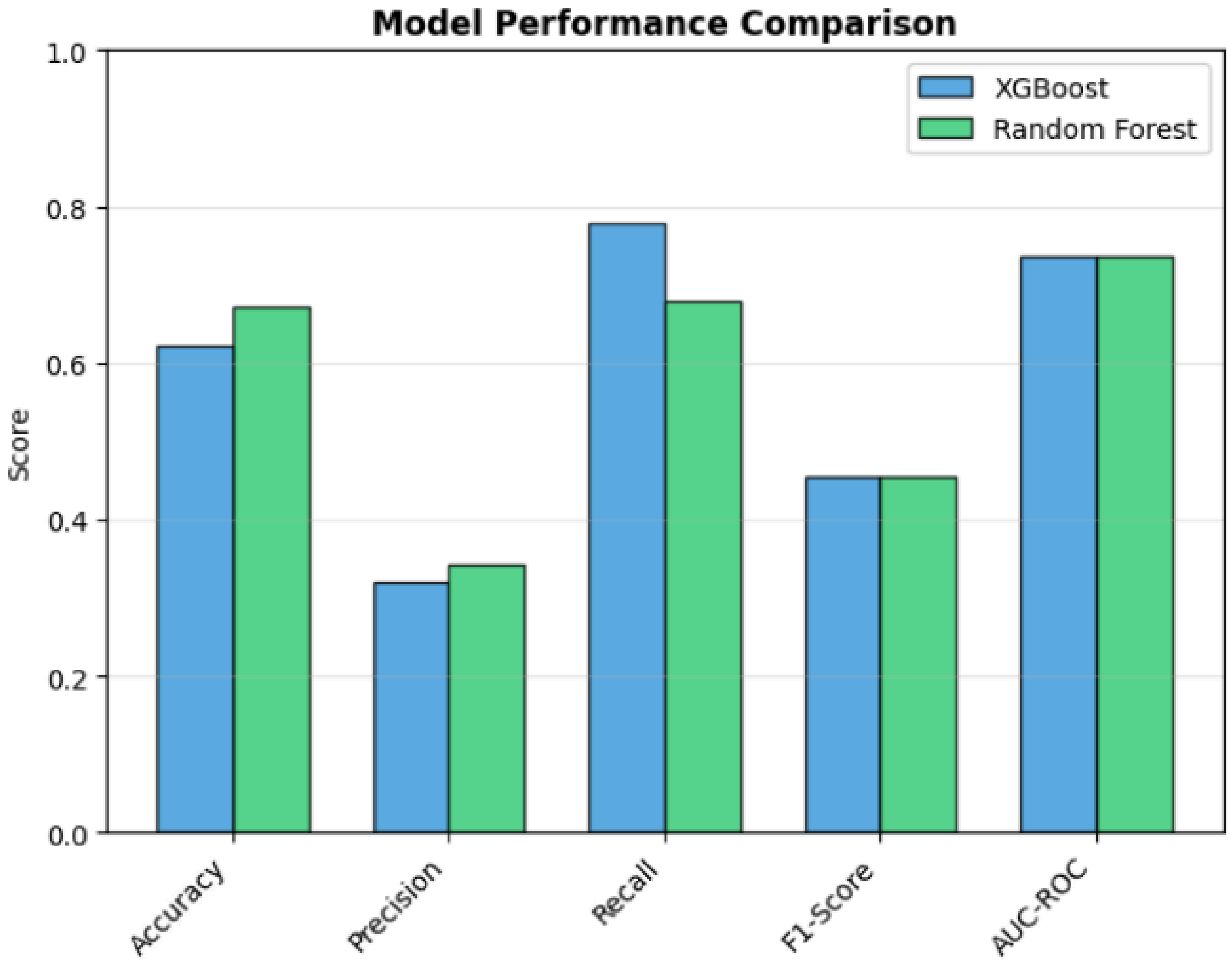
Random Forest Model

- The Random Forest model was trained on the prepared dataset using class weighting to address class imbalance.

5. RESULT & DICUSSION

Predictive Model Comparison: XGBoost vs Random Forest

- Both models were trained on the same preprocessed dataset and evaluated using multiple metrics.
- XGBoost achieved higher recall (76.7%) and AUC-ROC (0.738), making it more effective at identifying high-risk no-show patients.
- Random Forest performed slightly better in accuracy (67.1%) and precision (34.0%), but overall detection of no-shows was lower than XGBoost.
- Both models had similar F1-scores (~45%), showing balanced performance between precision and recall.
- Feature importance analysis highlighted AwaitingDays, WaitingCategory, and Age as the top predictors for both models.



6. CONCLUSION

This study analyzed hospital appointment data to understand and predict no-show behavior.PCA reduced numeric features, while clustering (K-means, Hierarchical, GMM) identified patient risk groups, with high-risk patients often having longer waiting times or missing SMS reminders.

Key Challenges

- Imbalanced data: Only ~20% no-show cases
- Mixed data types: Numeric, categorical, and time features
- Complex behavior: No clear pattern for some patients

Future Work

- Add external data (weather, distance, traffic)
- Explore advanced models (deep learning)
- Implement real-time no-show prediction
- Use personalized interventions for high-risk groups

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GITHUB LINK:

- HTTPS://GITHUB.COM/NATT2509/EDA_PROJECT/BLOB/MAIN/DATA.IPYNB

THANK YOU FOR YOUR ATTENTION

IF YOU HAVE ANY QUESTIONS CAN ASK !!!